```
import matplotlib.pyplot as plt
import numpy as np

# 1. DATA SETUP: All 4 specimens individually
specimens = [
    "Specimen 1\n(Gap 3.0 mm)",
    "Specimen 2\n(Gap 3.0 mm)",
    "Specimen 3\n(Gap 2.3 mm)",
    "Specimen 4\n(Gap 2.3 mm)",
]

# Relative Density (%) = 100% - Porosity (%)
# Gap 3.0 mm Theoretical Porosity = 73.68% -> Calculated Density = 26.32%
# Gap 2.3 mm Theoretical Porosity = 68.20% -> Calculated Density = 31.80%
calc_density = [26.32, 26.32, 31.80, 31.80]

# Apparent Porosities: 73.64%, 73.62%, 66.80%, 67.18%
# Apparent Densities: 26.36%, 26.38%, 33.20%, 32.82%
app_density = [100 - 73.64, 100 - 73.62, 100 - 66.80, 100 - 67.18]

# 2. PLOT SETUP
x = np.arange(len(specimens))
bar_width = 0.28

fig, ax = plt.subplots(figsize=(9, 5.5), dpi=300)
```

```
rects1 = ax.bar(
    x - bar_width / 2 - 0.02,
    calc_density,
    bar_width,
    label="Calculated / Theoretical Density",
    color="#2b5c8f",
    edgecolor="black",
    linewidth=0.8,
)

rects2 = ax.bar(
    x + bar_width / 2 + 0.02,
    app_density,
    bar_width,
    label="Apparent / Printed Density",
    color="#d95f02",
    edgecolor="black",
    linewidth=0.8,
)



# 3. VALUE LABELS ABOVE BARS
def add_labels(rects):
    for rect in rects:
```

```
        height = rect.get_height()
        ax.annotate(
            f"{height:.2f}%",
            xy=(rect.get_x() + rect.get_width() / 2, height),
            xytext=(0, 5),
            textcoords="offset points",
            ha="center",
            va="bottom",
            fontsize=9.5,
            fontweight="bold",
        )



add_labels(rects1)
add_labels(rects2)

# 4. TITLES & FORMATTING
title_text = "Calculated Density vs. Apparent Density (All 4 Specimens)"
subtitle_text = "Relative Density (% of solid material)"

ax.set_title(
    f"{title_text}\n({subtitle_text})", fontsize=11, fontweight="bold", pad=15
)
ax.set_ylabel("Relative Density (%)", fontsize=11, fontweight="bold")
ax.set_xticks(x)
```

```
ax.set_xticklabels(specimens, fontsize=9.5, fontweight="bold")
ax.set_ylim(0, 42)

# 5. LEGEND & GRID
ax.set_axisbelow(True)
ax.yaxis.grid(True, linestyle="--", alpha=0.5)
ax.legend(
    loc="upper center",
    bbox_to_anchor=(0.5, -0.15),
    ncol=2,
    frameon=True,
    fontsize=9.5,
)

plt.tight_layout()
plt.savefig("density_all_4_specimens.png", dpi=300, bbox_inches="tight")
plt.show()
```